STRhub Verified

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Technical Reproducibility Validation Report

STRsearch

Version c70179b
Autosomal STR

Verification date	2026-08-11
Repository commit	c70179b3b175adc82a7314409af06900b3861d61
Attestation level	**Available**
Permalink	https://strhub.app/verified/strsearch-c70179b
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**1/5 gates passed. Available**
ATTESTATION LEVEL	**Available**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRsearch
Version	c70179b
Repository	https://github.com/AnJingwd/STRsearch
Commit (pinned)	c70179b3b175adc82a7314409af06900b3861d61
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-08-11 12:39:10 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/31491770503

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRsearch c70179b · commit c70179b
$ mkdir \
-p \
/data/out \
&& \
python3 \
/opt/tool/pipeline.py \
from_bam \
--working_path \
/data/out \
--sample \
sample \
--sex \
male \
--bam \
/data/in/input.bam \
--ref_bed \
/data/in/regions.bed \
--genotypes \
/data/out/sample_genotypes.txt \
--multiple_alleles \
/data/out/sample_multiple_alleles.txt \
--qc_matrix \
/data/out/sample_qc_matrix.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	—
Runs	Executes end-to-end without crashing	—
Expected IO	Produces a non-empty file in the declared format	—
Output Structure Validation	Output structure matches expected genotype-bearing format	—
Execution log	strsearch-c70179b.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	sample_genotypes.txt
Format	TSV
Sequence records	Not assessed
Distinct STR loci	Not assessed
Total reads	Not assessed
Max depth (single locus)	Not assessed

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—
Regions BED	Provided by the tool author (covers 24 of 24 supported loci)

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

- **Runs end-to-end**

  STRsearch c70179b installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 1 verification gates were passed.

- **Produces valid output**

  The tool generated a structurally valid TSV output file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

- **No bundled demo data**

  STRsearch does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.